Supplementary Material

Risk-Calibrated Few-Shot Industrial Anomaly Detection with Human-in-the-Loop Inspection

This supplement reports exact calibration/test composition and complete normalized operating-cost sensitivity for the reported experiments. Machine-readable trial-level and aggregate CSV files are provided in the supplementary result directory included with the source package.

Table S1. Exact MVTec AD composition at $k = 4$

The support set contains four normal images. Calibration and test entries are normal/defective counts. Counts are identical across the five deterministic seeds; seed selection changes image identity, not the category totals.

Category	Normal support	Calibration	Held-out test
Bottle	4	67/18	158/45
Cable	4	83/27	195/65
Capsule	4	71/32	167/77
Carpet	4	91/26	213/63
Grid	4	84/17	197/40
Hazelnut	4	128/21	299/49
Leather	4	81/27	192/65
Metal Nut	4	71/27	167/66
Pill	4	86/42	203/99
Screw	4	107/35	250/84
Tile	4	77/25	182/59
Toothbrush	4	20/9	48/21
Transistor	4	80/12	189/28
Wood	4	78/18	184/42
Zipper	4	80/35	188/84

Table S2. Complete normalized operating-cost sensitivity

The normalized expected cost follows Eq. (5.2) of the manuscript. At each hypothetical defect prevalence π, manual-review cost is reweighted from the class-conditional manual-review rates, $\mathrm{MIR}_D = \Pr\{\text{Manual} \mid y = 1\}$ and $\mathrm{MIR}_N = \Pr\{\text{Manual} \mid y = 0\}$, rather than holding the benchmark class mixture fixed. The cost triples $(C_{\text{miss}}, C_{\text{false}}, C_{\text{manual}})$ are Balanced $(20, 5, 1)$, Safety-critical $(100, 5, 1)$, and Labor-expensive $(50, 5, 5)$. Costs are calculated per category-seed trial and then averaged. Values are dimensionless scenario costs, not monetary estimates.

Profile	Prev. (%)	RC-FS-IAD+	WinCLIP	DINOv2-PatchMemory	PatchCore-strong	SPADE-kNN-full	PatchCore-vanilla	PaDiM-diag
Balanced	1	0.950	**0.907**	1.039	0.965	1.000	1.063	1.146
Balanced	5	0.958	**0.927**	1.051	0.978	1.009	1.080	1.161
Balanced	10	0.969	**0.952**	1.067	0.993	1.020	1.100	1.180
Safety-critical	1	0.983	**0.950**	1.076	1.002	1.035	1.105	1.180
Safety-critical	5	**1.125**	1.145	1.236	1.162	1.184	1.287	1.330
Safety-critical	10	**1.302**	1.388	1.437	1.361	1.369	1.516	1.517
Labor-expensive	1	2.852	**2.563**	3.295	2.998	3.046	3.407	3.742
Labor-expensive	5	2.887	**2.632**	3.341	3.041	3.082	3.462	3.812
Labor-expensive	10	2.930	**2.720**	3.399	3.096	3.126	3.530	3.898

Machine-readable result files

The `results` directory contains the complete trial-level data for the nested VisA evaluation, k-shot sweep, controlled and modern baselines, expanded ablation, distribution-shift stress tests, human-review sensitivity, practical-significance calculations, and normalized cost analysis. The file `README_results.txt` describes the contents.